

Feature Detail & Demo Reference

This document is the **single source of truth** for implementing the CS2 RCON web interface (this repo is a fork of `nokit` / `defuse`). It captures everything observed in the public demo at <https://nokit.app/demo> so that developers **never need to visit the demo again**.

It is split into two parts:

1. **Feature Status Table** — a quick overview of what is actually implemented vs. what is currently only a static demo / planned.
2. **Detailed Feature Specifications** — an exhaustive, per-feature reference of every UI element, data field, column, interaction, chart, color code, status indicator, example value, and technical/implementation note recorded from the demo.

Demo versions observed: `nokit v0.1.0-rc.1` (footer), `relay v0.4.2` (sidebar footer). License MIT. Deploy target: Docker image `ghcr.io/nokit/nokit:latest`, default web port 8080, data volume `/data`, single ~14 MB Go binary running alongside `srcds` over loopback RCON.

Implementation will begin with the **Dashboard**.

Section 1 — Feature Status Table

Feature	Status	Notes
Login / Auth	Implemented	Password-based login, session tokens in SQLite
Multi-server switcher	Implemented	Add/remove servers (name, RCON host, password) via header

Feature	Status	Notes
RCON Console	Implemented	Full terminal UI: timestamps, color-coded output, ↑↓ history, Ctrl+L clear, Tab autocomplete (5000+ CS2 CVARs/commands), RCON macros sidebar with localStorage persistence, clickable history panel, live/paused scroll toggle, Copy session & Export
Dashboard	Implemented	Live stat cards (CPU/tick/players) with sparklines, server status, quick actions, round info, recent output — polls status/stats over RCON every 6s
Live Logs	Not built	Demo only / planned
Players	Not built	Demo only / planned
Maps	Implemented	Standard map pool (12 maps), favorites system (localStorage), workshop map support, map cycle editor, RCON integration (changelevel, host_workshop_map)
CVAR Presets	Not built	Demo only / planned

Feature	Status	Notes
Config Editor	Implemented	File browser tree view, code editor with line numbers, unsaved changes tracking, save/reload/exec via RCON, support for .cfg and .json files
Plugins	Not built	Demo only / planned
Scheduler	Not built	Demo only / planned
Admin	Not built	Demo only / planned

Section 2 — Detailed Feature Specifications

Global Layout & Chrome (present on all pages)

These elements are shared across every page and should be built as part of the app shell.

Top navigation bar (full width, dark) **Left cluster:** - **nokit** logo (green square “n” icon + wordmark) — top-left. - **Server switcher dropdown:** label `fra1.mm`, subtext `5.39.42.118:27015`, with a small green online dot and a chevron (dropdown to switch servers / add-remove servers). See §11. - **tickrate** indicator: `127.8` - **players** indicator: `9/12` - **map** indicator: `de_mirage`

Right cluster: - **Search box:** placeholder `cvars`, `players...` with a `K / K` keyboard-shortcut hint badge (command palette style search). See §13. - Small **HK / K** badge button. - **External-link icon** (open server / launch). - **Settings gear icon.** - **User menu:** avatar + `m0use@` (current user), with dropdown. - **Power / logout icon** (far right).

Left sidebar navigation (grouped)

- **SERVER**
 - Dashboard (home icon) — active/highlighted green
 - RCON Console — has a small `K/HK` badge on the right
 - Live Logs
 - Players
- **CONFIGURATION**
 - Maps
 - CVAR Presets
 - Config Editor
- **SYSTEM**

- Plugins — has an amber/orange numeric badge 1 (e.g. update/notification count)
- Scheduler
- Admin

Sidebar footer: relay v0.4.2, a throughput indicator ↑ 2.1 kb/s · ↓ 18 kb/s, and a green WS (WebSocket connected) status.

Page footer (bottom, full width)

- Left: nokit v0.1.0-rc.1 · MIT · github · docs · changelog
- Right (status strip): srcds OK · relay OK · 10/07/2026 · 16:16:14 · Europe/Berlin (UTC+2) — shows service health, date, live clock, timezone. The footer clock ticks live (observed 16:16 → 16:30 across the session).

1. Dashboard

Page title: **Dashboard**. Subtitle: fra1.mm · 5.39.42.118:27015 · connected via rcon. Top-right of page: green live pill + Refresh button.

Top stat cards (4 across)

Card	Big value	Sub-line
PLAYERS	9 / 12 (green +2 delta badge)	last 5m
TICK RATE	127.8 hz	target 128
CPU	22 %	4-core · 0.88 load avg
RAM	2.1 / 8 GB	resident

Current round panel (left, ~2/3 width) Header: **Current round** — right side: tick #88669 · uptime 11h 02m. - Map thumbnail image with label de_mirage and a competitive badge overlay. - Field rows (label → value): - map → de_mirage (active pool) - gamemode → competitive · MR12 + OT - round → 8 / 24 - CT 4 : 3 T - round time → 0:47 remaining - warmup → ended · live - session → 2h 14m · matchzy "live"

Quick actions panel (right, ~1/3 width) Header: **Quick actions**. Buttons top-to-bottom: - Restart server - Segmented toggle: [match | practice | pug] — match selected (highlighted). - No password - End warmup - Auto-shuffle teams - Stop server (red/danger styling)

Mini metric chart widgets (4 across, sparkline/area charts)

Widget	Range	Value	Chart color	Footer
Players	24h	9 / 12	green line	peak 12 · avg 6.4
CPU	5m	22%	blue area	peak 41% · sustained 28%
Memory	5m	2.1 GB	purple line	26% of 8 GB
Tick	5m	127.8 hz	orange line	target 128 · jitter 0.4

stats // rcon dump panel (bottom-left) Header: **stats // rcon dump** — right: refreshed 4s ago. Tabular columns: CPU | NetIn | NetOut | Uptime | Maps | FPS | Players | Svms | +-ms | ~tick Row values: 22.0 | 18.4 | 61.7 | 11h02m | 14 | 127.8 | 9 | 7.42 | 0.40 | 127.8 Below the table (key/value lines): - version: 1.40.6.3 (cs2) - os: linux x86_64 (debian 12) - gotv: 1x spec on :27020 - protocol: v15 / build 9462 - vac: secure - workshop: 16 maps · 281 MB

Recent events panel (bottom-right) Header: **Recent events** — right: view all → link. Event rows (timestamp + actor + action): - 12:51 m0use@ ended warmup - 12:49 jaeger kicked h4rddy_woods - afk - 12:40 m0use@ changed map: de_mirage - 12:38 jaeger applied preset: MR12 (24+0T) - 12:35 m0use@ edited gamemode_competitive.cfg - 12:18 system GotvFix: bind failure

Implementation notes: dashboard/stats auto-refresh every ~4s. The stat cards and mini charts are driven by the **status** RCON dump plus a metrics history buffer. The Recent events panel mirrors the Admin → Audit log.

Current implementation status (this fork): The Dashboard is built and is the default view when a server is selected (`web/src/pages/dashboard-page.tsx`), with a Dashboard/Console tab bar in `server-page.tsx`. It polls the **status** and **stats** RCON commands every 6s via the `useServerStats` hook (`web/src/hooks/useServerStats.ts`), parsing them with `web/src/lib/rcon-parse.ts` (tolerant of multiple CS2 output formats). - **Stat cards** (`StatsCard` + `Sparkline`): Players (green), Tick rate (orange, target 128), CPU (blue area chart, with FPS sub-line), and RAM. Sparklines are rendered from a rolling in-memory history. RAM is shown as **n/a** because CS2's `stats/status` do not report server memory over RCON. - **Server status** (`ServerStatus`): online/offline dot, name, address, map, hostname, players, uptime, version, OS — all parsed from `status/stats`. - **Quick actions** (`QuickActions`): Restart server (`_restart`, confirmed), match/practice/pug segmented toggle (`exec <mode>`), clear password (`sv_password ""`), End warmup (`mp_warmup_end`), Auto-shuffle teams (`mp_scambleteams 1`), and Stop server (`quit`, confirmed) — each sends the command

over the existing `/api/servers/:id/rcon` endpoint and surfaces a success/error line. - **Round info** (`RoundInfo`): shows map, phase (live/offline) and player count from `status`; round number/score/timer are shown as - with a note that they will come from the Live Logs stream (not exposed by `status/stats`). - **Recent output** (`RecentOutput`): shows the last lines of the most recent `status` poll and a link that switches to the full Console tab.

2. RCON Console

Page title: **RCON Console**. Subtitle: `tab` to autocomplete · `↑↓` history · `ctrl-l` clears. Top-right controls: `[ live | paused ]` segmented toggle (live selected), `Copy session` button, `Export` button.

Console output area (left, large monospace terminal, ~2/3 width)
 Green/amber terminal text. Commands echoed with timestamp prefix `rcon HH:MM:SS >`. Command values are syntax-highlighted (numbers/limits in color). Example session content: - `rcon 12:50:00 > status` - `hostname: nokit | fra1.mm · map: de_mirage · players: 10/12 · pubkey: 8f12...ed94` - `rcon 12:52:02 > mp_warmuptime 15` - `"mp_warmuptime" = "15" ( def. "30" min. 0.000000 max. 90.000000 ) replicated` - `Time after a new round...` - `rcon 12:54:04 > sv_password` - `"sv_password" = "*****"` `archive notify` - `Password to join the server.` - `rcon 12:56:06 > mp_warmup_end` - `[matchzy] warmup ended. live in 5...` - - - `live` - -

Input bar (bottom of console) Prompt: `rcon` monospace input field, pre-filled demo text `mp_war`. Has a small submit/enter affordance on the right.

Quirk: in the demo the input is static — typing appends but no live autocomplete dropdown fires. The subtitle advertises `tab-to-autocomplete`, `↑↓` history, and `ctrl-l` to clear as intended behaviors.

RCON macros panel (right sidebar) Header: **RCON macros**. Each macro is a button showing a friendly label + the underlying command (greyed monospace): `| Label | Command | |——-|———|` `| warmup → live` `| mp_warmuptime 15 ; m...` `| | reset score | mp_restartgame 1 |` `| | knife → side pick | matchzy_knife |` `| | pause match | matchzy_pause |` `| | kick all bots | bot_kick |` `| | demo: start | tv_record demos/manu...` `| | demo: stop | tv_stoprecord |` `| | fix stuck round | mp_unpause_match ; m...` `| - + New macro...` button at bottom (create custom macro).

History panel (right sidebar, below macros) Header: **History**. Clickable recent commands list (re-run on click): - `status` - `mp_warmuptime`

15 - sv_password - mp_warmup_end - changelevel de_inferno - tv_record
demos/2026-05-21

Current implementation status (this fork): The RCON Console is fully built as a real terminal UI. The component lives in `web/src/components/console.tsx` (exported as `Console`) and is rendered from the Console tab in `web/src/pages/server-page.tsx`. It uses the existing `POST /api/servers/:id/rcon` endpoint — no new backend routes were added. - **Layout:** two-column split — a left terminal area (~65% width) and a right sidebar (~35% width). - **Terminal output:** every command is echoed with a timestamp prefix `rcon HH:MM:SS >` and output is color-coded — green for the command prompt/echo, amber for CVAR values, red for errors, and white/muted for regular output. - **Input bar:** monospace prompt `rcon` with shell-style keyboard shortcuts — `↑↓` navigate command history, `Ctrl+L` clears the terminal, and `Tab` triggers autocomplete with a dropdown covering **5000+ CS2 CVARs/commands** (loaded from `web/src/data/cs2-commands.json`, sourced from ArminC-CS2-Cvars). The autocomplete is capped at 50 suggestions per prefix for performance. - **Live/paused scroll toggle:** in live mode the view auto-snaps to the newest line; paused mode lets the user scroll freely through the backlog without being pulled back to the bottom. - **Session tools:** `Copy session` copies the full session log to the clipboard, and `Export` downloads it as a `.txt` file. - **RCON macros panel:** 8 built-in macros (warmup → live, reset score, knife round, pause match, kick bots, demo start, demo stop, fix stuck round) plus custom macro creation via the `+ New macro` form. All custom macros are persisted in `localStorage` under the key `nokit_console_macros`. - **History panel:** shows the last 20 unique commands (most recent first); clicking an entry pastes it into the input. History is persisted across reloads in `localStorage` under the key `nokit_console_history`.

3. Live Logs

Page title: **Live Logs**. Subtitle: `srcds_logaddress` · SSE to `nokit-relay` · 19 retained · 19 shown (shown count updates when filtering). Top-right controls: green **LIVE** pill, **Pause** button, **Download .log** button.

Filter tabs (with count badges) All 19 · Chat 3 · Kills 4 · Rounds 5 · Connect / team 2 · Plugin 1 - **Functional:** clicking a tab filters the stream by event category and updates the “N shown” count (verified — Chat tab showed only the 3 chat lines).

Search / grep bar

- Input with placeholder: `grep - e.g. STEAM_1:0, "awp", defused and a / regex toggle affordance at the right.`
- Toggle checkbox: **hide rcon noise**.
- Far right: `tail -f · 1.4k lines/min` throughput label.

Log stream (monospace, color-coded by event type)

Each line: HH:MM:SS <TYPE> message. The event-type keyword is colored. Types seen:

rcon, world, round, kill, chat, conn, team, srv, plug. Example lines: -

12:51:08 rcon > mp_warmup_end - 12:51:08 world World triggered

"Warmup_End" - 12:51:09 round World triggered "Round_Start"

- 12:51:14 kill "m0use<2><STEAM_1:0:48211904><CT>" killed

"zywoo<7><STEAM_1:1:11923844><T>" with "ak47" (headshot) - 12:51:23

chat "m0use<2><STEAM_1:0:48211904><CT>" say "gg wp" - 12:51:25

kill "NEKO<5><STEAM_1:1:78231004><T>" killed "ferr<3><STEAM_1:1:91134472><CT>"

with "awp" - 12:51:27 round Team "T" triggered "SFUI_Notice_Target_Saved"

(CT "1") (T "0") - 12:51:27 round World triggered "Round_End" -

12:51:32 conn "jL<9><STEAM_1:0:90120431>>" connected, address

"78.34.221.41:27015" - 12:51:34 team "jL<9><STEAM_1:0:90120431><Unassigned>"

switched from team <Unassigned> to <TERRORIST> - 12:51:40 chat

"NEKO<5><STEAM_1:1:78231004><T>" say "rotate B" - 12:51:55 rcon

> status - 12:51:55 srv hostname: nokit | fra1.mm · map:

de_mirage · players: 10/12 - 12:52:01 kill "twist<6><STEAM_1:1:12349803><CT>"

killed "byomas<8><STEAM_1:1:43712001><T>" with "deagle" (headshot)

- 12:52:08 chat "olof<4><STEAM_1:0:55102941><CT>" say_team

"info one toxic at long" - 12:52:12 plug [WeaponPaints] failed

to fetch skin metadata for STEAM_1:1:43712001 (timeout 5s) -

12:52:30 kill "zywoo<7><STEAM_1:1:11923844><T>" killed "m0use<2><STEAM_1:0:48211904><CT>"

with "awp" (penetrated) - 12:52:55 round Team "CT" triggered

"SFUI_Notice_Bomb_Defused" (CT "1") (T "1") - 12:53:01 rcon >

changelevel de_inferno

Stream footer tail awaiting next event · backpressure 0 · last frame 1.2s ago with a green **SSE** connection indicator (right).

Implementation notes: logs are ingested via `srcds_logaddress` and streamed to the browser over **Server-Sent Events (SSE)** through nokit-relay. The stream shows backpressure and last-frame timing.

4. Players

Page title: **Players**. Subtitle: auto-refresh every 4s · 10 online · 4 on watchlist · 4 bans. Top-right buttons: Auto-shuffle, Mass kick...

green + Add to watchlist.

Tabs (with count badges) Live 10 · Watchlist 4 · Bans 4 · Session history

Common controls

- Search input: placeholder `search name, steamid, country, notes`.
- Team filter segmented toggle: [all | CT | T | spec] (all selected).
- Filter button (right).

Live tab — **table columns** NAME | STEAMID | TEAM | PING | TIME | K/D | COUNTRY - **NAME**: some rows have an amber flag icon + a green watch badge (on watchlist). Rows with high ping shown in a warning color. - **TEAM**: colored pill — CT (blue) / T (amber/orange). - **PING**: e.g. 18ms (high ping like 86ms highlighted). - **TIME**: connected time e.g. 1:24:11. - **K/D**: ratio e.g. 1.31. - **COUNTRY**: flag icon + 2-letter code.

NAME	STEAMID	TEAM	PING	TIME	K/D	COUNTRY	
m0use_ (watch)	STEAM_1:0:48211904	CT	1904	18ms	1:24:11	1.31	SE
ferr	STEAM_1:1:90134472	CT	4472	22ms	1:24:08	0.94	DE
kqly.exe	STEAM_1:0:32220110	CT	20110	41ms	1:18:54	1.07	PL
twist	STEAM_1:1:12349803	CT	9803	24ms	1:17:02	1.18	SE
olof	STEAM_1:0:56102941	CT	2941	28ms	1:11:33	0.88	SE
NEKO	STEAM_1:1:78231004	CT	1004	19ms	1:24:11	1.42	JP
zywoo*	STEAM_1:0:1923844	CT	23844	26ms	1:23:01	1.95	FR
bymas	STEAM_1:1:43712001	CT	12001	33ms	0:58:21	1.04	LT
jL	STEAM_1:0:90120431	CT	20431	29ms	0:42:11	1.08	FI
h4rdy_woods (watch)	STEAM_1:1:77124930	CT	124930	86ms	0:11:54	0.73	BR

Row hover/click reveals inline action buttons (right side of row): profile, spectate, kick, ban... (red/danger).

Watchlist tab — **table columns** NAME | STEAMID | REASON | STATUS |
NAME | STEAMID | REASON | STATUS | | | | | | m0use_
| STEAM_1:0:48211904 | wallhack reports x4 | online · this server (green) | |
h4rdy_woods | STEAM_1:1:77124930 | griefing — voted twice | online · this
server (green) | | silent.knife | STEAM_1:1:20300912 | team-damage spammer |
offline (grey) | | badboy_3000 | STEAM_1:0:65332100 | racism — 1mo warn
| offline (grey) | - Each row has an amber flag icon. STATUS is a colored pill
(green online / grey offline).

Bans tab — **table columns** NAME | STEAMID | REASON | BY | EXPIRES
 | NAME | STEAMID | REASON | BY | EXPIRES | |——|——|——|——|——
 ——| | lobotomy | STEAM_1:0:55512090 | wallhack — demo evidence | m0use@
 | 2026-06-12 18:00 (amber) | | rage_quitter | STEAM_1:1:11203044 | griefing |
 admin | 2026-05-24 04:00 (amber) | | idiot | STEAM_1:0:99687700 | racism —
 auto-detect | system | permanent (red) | | spinbot | STEAM_1:1:00091123 | aim
 hack — confirmed demo | m0use@ | permanent (red) | - EXPIRES: temporary
 bans show a date pill (amber); permanent bans show a red **permanent** pill.

Session history tab Empty state message: No session history pruned yet. Last 30d retained.

Implementation notes: player list auto-refreshes every ~4s (parsed from **status**). Watchlist and bans are persisted server-side.

5. Maps

Page title: **Maps**. Subtitle: 12 standard · 4 workshop · current: de_mirage. Top-right buttons: Sync workshop, Browse collection.

FAVORITES row Section label FAVORITES with a show favs only toggle (right). Compact favorite cards (thumbnail + name + mode badge): - Mirage (active badge) · comp - Inferno · comp - Dust II · comp - Overpass · comp - aim_botz · practice

STANDARD POOL grid (4-across cards) Each card: map thumbnail (radar-style with A/B bomb site markers), map name, a favorite star, and a mode badge (comp, hostage, practice). The active map has a green active badge and highlighted green border. Maps: de_mirage (active), de_inferno, de_dust2, de_nuke, de_overpass, de_anubis, de_vertigo, de_ancient, cs_office (hostage), cs_italy (hostage), de_train (comp), aim_botz (practice).

Map change interaction (verified): clicking a map card selects it as active — the green active badge and highlight move to the clicked card (both its pool card and favorites card update). This is how you change the current map.

WORKSHOP section Label: WORKSHOP 281 MB CACHED · /HOME/CS2/SERVER/CSGO/MAPS/WORKSHOP.
 - Workshop ID input: placeholder workshop ID - e.g. 3070900859. - host_workshop_map button. - Green Download & switch button.

Workshop map cards (thumbnail + name + author + subscriber count + workshop ID +): | Name | Author | Subs | Workshop ID | |——|——|——|——|——
 ——| | de_cache_redux | by shawn | 281k subs | #3070900859 | | awp_lego |

by 3klikphilip | 92k subs | #3070192884 | | surf_kitsune_ksf | by kitsune | 14k subs | #3071182234 | | de_season | by sumamon | 441k subs | #3072201991 |

MAP CYCLE editor (bottom) Label MAP CYCLE. Ordered chip list (each chip has a numeric order + name + × remove): order: 01 de_mirage × · 02 de_inferno × · 03 de_nuke × · 04 de_dust2 × · 05 de_anubis × · 06 de_overpass × · 07 de_ancient × - + add button to append a map. - Right: save mapcycle.txt button (writes the cycle to mapcycle.txt).

Current implementation status (this fork): The Maps panel is fully built as a functional page. The component lives in `web/src/pages/maps-page.tsx` (exported as `MapsPage`) and is rendered from the `/servers/:id/maps` route in `App.tsx`. It uses the existing `POST /api/servers/:id/rcon` endpoint for map changes — no new backend routes were added. - **Standard map pool:** 12 CS2 maps loaded from `web/src/data/cs2-maps.json` (Mirage, Inferno, Dust II, Nuke, Overpass, Anubis, Vertigo, Ancient, Train, Office, Italy, aim_botz) with categorization by mode (comp, hostage, practice). - **Favorites system:** Users can star/unstar maps; favorites are persisted in `localStorage` under `nokit_map_favorites`. A “show favs only” toggle filters the grid to show only favorited maps. Favorite maps are displayed in a compact horizontal row at the top for quick access. - **Active map tracking:** The currently active map is highlighted with a green border and “active” badge. Map changes are triggered via `RCON changelevel <mapid>` command. - **Workshop maps:** Demo section displays 4 workshop maps (`de_cache_redux`, `awp_lego`, `surf_kitsune_ksf`, `de_season`) with author, subscriber count, and workshop ID. Users can input a workshop ID and load it via `RCON host_workshop_map <workshopid>` command. - **Map cycle editor:** Displays an ordered list of maps in the rotation (persisted in `localStorage` under `nokit_map_cycle`). Users can remove maps from the cycle; the default cycle includes 7 competitive maps. “Save mapcycle.txt” button is ready for backend integration to write the cycle to the server’s `mapcycle.txt` file. - **Responsive grid:** Map cards display in a 4-column grid (responsive: 1 col mobile, 2 col tablet, 3 col desktop, 4 col wide) with hover states and click-to-change functionality. - **TODO:** Map thumbnail images (currently showing placeholder gradients), drag-and-drop reordering for map cycle, Steam Workshop API integration for fetching workshop map metadata and thumbnails.

6. CVAR Presets

Page title: **CVAR Presets**. Subtitle: 7 applied · changes preview as inline rcon · server.cfg untouched until "Persist". Top-right buttons: + New custom preset, Export .cfg, green Apply all selected.

Layout Left ~2/3: preset categories, each a bordered group with a header CATEGORY N / M APPLIED and a row of toggle chips (checkbox pill; selected = green fill + checkmark). Each group ends with a + custom... chip to add a custom preset. Right ~1/3: **Inline preview** panel + **Quick switch** panel.

Preset categories & chips

- **ROUND SETTINGS 2 / 6 APPLIED:** MR12 (24+OT) · MR15 (30) · Short (B09) · Warmup 15s · Warmup 60s · Freeze 10s · + custom...
- **ECONOMY 2 / 4 APPLIED:** Comp default · Pistol only · Rich start · 1:55 round · + custom...
- **MOVEMENT 1 / 3 APPLIED:** Default · Surf · KZ · + custom...
- **SERVER 2 / 4 APPLIED:** 128 tick · 64 tick · LAN · Pure 2 · + custom...
- **PRACTICE 0 / 4 APPLIED:** Nade practice · Infinite nades · No-flash bots · Auto-respawn · + custom...

Inline preview panel (right) Header: **Inline preview** — right: 7 commands (count updates live). Monospace numbered listing grouped by preset comment header: - #01 # MR12 (24+OT) → mp_maxrounds 24 ; mp_overtime_enable 1 - #02 # Warmup 15s → mp_warmuptime 15 - #03 # Comp default → mp_startmoney 800 ; mp_maxmoney 16000 - #04 # 1:55 round → mp_roundtime 1.92 - #05 # Default → sv_accelerate 5.5 ; sv_airaccelerate 12 - #06 # 128 tick → sv_minrate 128000 ; sv_maxupdaterate 128 Footer: = to apply & persist + green **Apply & persist** button.

Interaction (verified): toggling a preset chip live-updates the inline preview — enabling MR15 (30) inserted a #02 # MR15 (30) → mp_maxrounds 30 ; mp_overtime_enable 0 block and the command counter incremented 7 → 8.

Quick switch panel (right, below preview) Header: **Quick switch**. Full scenario presets (each with a keyboard shortcut badge): - Match - 5v5 MR12 + OT — green active badge - Practice - solo nades — 1 - Pug - public retake — 2 - LAN - sv_lan 1, no auth — 3

Implementation notes: presets preview as inline RCON commands; server.cfg is untouched until the user hits **Persist**. Applying sends the commands over RCON; persisting writes them to the config.

7. Config Editor

Page title: **Config Editor**. Subtitle (breadcrumb path): /home/cs2/server/csgo/cfg/gamemodes/gamemode.
Top-right: amber unsaved badge, Reload from disk button, >_ exec via rcon button, green Save & apply button.

File browser (left panel, tree) Header: cfg / with a + (new file) button. Tree entries show a “last modified” age on the right: - autoexec.cfg — 2d - server.cfg — 1h - banned_users.cfg — 3h - banned_ip.cfg — 7d - gamemodes (folder, 4) - gamemode_competitive.cfg — 12m - gamemode_competitive_server.cfg — (open/selected) - gamemode_casual.cfg - gamemode_deathmatch.cfg — 14d - matchzy (folder, 3) - mr12_competitive.cfg — 8h - live.cfg — 0h - warmup.cfg — 0h - cs_sharp (folder, 2) - WeaponPaints.json — 5d - MatchZy.json — 8h

Editor (right panel — code editor with line numbers) Top status strip (right): utf-8 · LF · 80 lines · last edit 12m ago by m0use@. Syntax-highlighted .cfg content with line numbers. Comments in green (// ...), cvar names one color, values another. Sample content:

```
1 // gamemode_competitive_server.cfg
2 // matchzy + tournament defaults - fra1.mm
3 // last edit: m0use@ - 12:35
5 mp_maxrounds           24
6 mp_overtime_enable     1
7 mp_overtime_maxrounds  6
8 mp_overtime_startmoney 16000
9 mp_freezetime          15
10 mp_roundtime           1.92
11 mp_warmuptime          15
12 mp_startmoney          800
13 mp_maxmoney            16000
14 mp_buy_anywhere        0
15 mp_team_intro_period   0
17 sv_pure                2
18 sv_minrate             128000
19 sv_maxupdaterate       128
20 sv_minupdaterate       128
21 sv_cheats              0
23 // matchzy
24 matchzy_minimum_ready_required 5
25 matchzy_kick_when_no_match_loaded 0
26 matchzy_demo_path      "matchzy_demos"
27 matchzy_chat_prefix     "[ nokit ]"
```

Quirk: in the demo the editor content is static — selecting another file in the tree does not swap the shown buffer. In the real implementation, selecting a file loads its buffer; the **unsaved** badge indicates dirty state; **Save & apply** writes to disk and can **exec via rcon**.

Current implementation status (this fork): The Config Editor is fully built as a functional page. The component lives in `web/src/pages/config-editor-page.tsx` (exported as `ConfigEditorPage`) and is rendered from the `/servers/:id/config` route in `App.tsx`. - **File browser tree:** Left sidebar (256px) with hierarchical tree view of config files. Mock data includes 7 files + 3 folders (gamemodes, matchzy, cs_sharp) with expand/collapse functionality. Each file shows “last modified” timestamp. - **Code editor:** Full-width textarea with line numbers, monospace font, dark theme (bg-zinc-950). Supports multi-line editing with proper line height (1.5rem). - **Unsaved changes tracking:** Compares current content with original content; shows amber “unsaved” badge when dirty. “Reload from disk” button resets to original content. - **File operations:** - **Save & apply:** Saves file to server filesystem (TODO: backend integration) - **Reload from disk:** Discards changes and reloads original content - **exec via rcon:** Executes the file via RCON `exec <filename>` command - **Header breadcrumb:** Shows full path `/home/cs2/server/csgo{selectedPath}` - **Footer status:** Displays encoding (utf-8), line endings (LF), line count, and last edit metadata - **File selection:** Click any file in tree to load its content (currently shows demo content for all files) - **TODO:** Real file loading from server, syntax highlighting for `.cfg/.json`, create new file functionality, backend integration for saving files to server filesystem

8. Plugins

Page title: **Plugins**. Subtitle: 8 installed · 1 update available · 1 error · 1 warning · CS# 1.0.305 + Metamod 2.0.0-12. Top-right buttons: Reload all, Update all (1), green + Browse registry.

Tabs (count badges) Installed 8 · Browse 24+ · Error log 1 (Error log badge is red).

Installed tab Controls: filter input filter installed - name, author, tag; segmented status filter [all | enabled | disabled | has update] (all selected); Install from .dll button (right).

Row layout: letter-avatar icon · name · version · status badge · tag

badge(s) · author @handle · updated <age> · category · install path /addons/cs_sharp/plugins/<x> · description line · then on the right: an **enable/disable toggle switch** + **Edit config** + **View logs** + **Details** + red trash/delete icon.

Status badges: green **enabled**, grey **disabled**, amber **warn**, red **failed**, blue **core**, amber **update** → x.y.z.

Installed plugins: | Plugin | Ver | Status | Author | Updated | Category |
 Path | Description | |-----|-----|-----|-----|-----|-----|
 CounterStrikeSharp | v1.0.305 | enabled, core | @roflmuffin | 12 May | core |
 /addons/cs_sharp/plugins/css | C# scripting host for CS2 dedicated servers.
 Loads .NET 8 plugin assemblies and exposes the gameserver event bus, RCON,
 and cvar APIs to managed code. | | Metamod:Source | v2.0.0-12 | enabled, core
 | @AlliedModders | 08 May | core | /addons/cs_sharp/plugins/mm | Source
 engine plugin loader. Required by CounterStrikeSharp. | | MatchZy | v0.8.7
 | enabled, update → 0.8.8 | @shobhit-pathak | 3d ago | competitive | /ad-
 dons/cs_sharp/plugins/matchzy | Pug, scrim, and tournament match man-
 agement with knife rounds, tactical pauses, and per-round demos. | | Kit-
 suneLab.MovementApi | v1.4.0 | enabled | @kitsunelab | 11d ago | utilities | /ad-
 dons/cs_sharp/plugins/mov | Movement hook API used by surf, kz, and bhop
 plugins. Provides tick-accurate per-player movement callbacks. | | Weapon-
 Paints | v4.0 | warn | @Nereziel | 5d ago | fun | /addons/cs_sharp/plugins/paints
 | Skin / glove / knife chooser. Persists selection per SteamID via MySQL.
 | | CS2-SimpleAdmin | v1.6.1 | enabled | @daffyy | 1mo ago | admin | /ad-
 dons/cs_sharp/plugins/sa | In-game admin commands — kick, ban, gag, mute,
 freeze, slay. Compatible with SourceMod admins.cfg. | | CallAdmin | v0.6.2 |
 disabled | @rumblefrog | 2mo ago | admin | /addons/cs_sharp/plugins/cadm |
 In-game report → Discord webhook with player context attached. | | GotvFix
 | v0.3.0 | failed | @whisper | 14d ago | utilities | /addons/cs_sharp/plugins/gotv
 | TCP relay shim for GOTV — works around Valve's GOTV2 broken multicast
 on Linux 6.x kernels. |

- The **failed** GotvFix row shows an inline red error box: [ERR]
 bind(0.0.0.0:27020) → EADDRINUSE - gotv relay disabled.

Browse tab (plugin registry / marketplace) Left rail: - **CATE-
 GORIES:** All 24 · Admin tools 4 · Competitive 4 · Fun 4 · Utilities 4 ·
 Maps 3 · Gamemodes 5. - **FILTERS** (checkboxes): Has screenshots · CS2
 compatible (checked) · MIT / Apache · No DB required. - **SOURCES:**
 Registry 327 (active) · Custom git 3 · Local .dll. Top bar: search **search** 327
 plugins - **name**, **author** (with / regex); sort options [**trending** | **most**
installed | **recently updated** | **name**] (trending selected).

FEATURED cards (icon · name · author · version · badge **featured/installed**
 · description · ↓ **downloads** · <age> · green **install** button): - CS2-
 RetakesAllocator v2.1.0 @b3none · featured · “Full retakes implementation —

bombsite rotation, automatic bomb planter, weapon allocator with...” · ↓47.4k · 3d ago · [install] - MatchZy v0.8.7 @shobhit-pathak · installed · “Pug, scrim & tournament match management with knife round, pauses, demos.” · ↓92.1k · 3d ago - WeaponPaints v4.0 @Nereziel · installed · “Skin / glove / knife chooser with web profile editor. MySQL-backed.” · ↓88.0k · 5d ago - Surf Timer v1.0.4 @kitsunelab · featured · “Per-map records, leaderboards, and per-zone splits with a TAS-quality replay bot.” · ↓21.6k · 4d ago · [install]

ALL PLUGINS · 24 (row: icon · name · version · author · ↓downloads · updated · category · tag chips · **Install** button + external-link icon): - AccessChecker v1.1.0 @j00nfist · ↓24.6k · 5d ago · Admin tools · #groups #auth · “Steam group + flag-based access control. Verifies via Steam Web API on connect with 5-minute cache.” - VoteManager v2.0.1 @riffri · ↓18.2k · 2w ago · Admin tools · #voting · “Veto, map vote, mute vote, kick vote. Configurable thresholds and player-side menus.” - SpectatorTools v0.9.4 @b3none · ↓9.7k · 1mo ago · Admin tools · #debug · “Force spectate, freeze, no-clip, and god-mode admin tools.”

Error log tab Controls: search search plugin output - message, plugin name, level; segmented level filter [all | errors | warnings | info]; Export button (right). Monospace log lines YYYY-MM-DD HH:MM:SS [LEVEL] PluginName - message. Levels color-coded: [WARN] amber, [ERR] red, [INFO] grey/blue. - 2026-05-21 12:48:02 [WARN] WeaponPaints - failed to fetch skin metadata for STEAM_1:1:43712001 (timeout 5s) - 2026-05-21 12:18:54 [ERR] GotvFix - bind(0.0.0.0:27020) → EADDRINUSE - gotv relay disabled - 2026-05-21 11:30:11 [INFO] MatchZy - matchzy: live config "mr12_competitive.cfg" applied - 2026-05-21 11:08:31 [INFO] CounterStrikeSharp - plugin host started · 8 plugins loaded in 412ms - 2026-05-21 10:55:14 [INFO] MatchZy - autoloader: warmup.cfg → live.cfg transition configured - 2026-05-21 10:42:11 [INFO] WeaponPaints - mysql: connected to 127.0.0.1:3306 (db=weapons, latency 1.4ms) - 2026-05-21 10:39:02 [WARN] GotvFix - falling back to TCP relay - kernel multicast probe failed - 2026-05-21 10:38:54 [ERR] GotvFix - unable to subscribe to GotvFix.RelayPort (27020) - retrying in 30s Footer: plugin-log last 24h · 1 err · 2 warn · 12 info · 8 lines + clear button.

9. Scheduler

Page title: **Scheduler**. Subtitle: 4/5 active · system tz: Europe/Berlin · cron via nokit-relay. Top-right buttons: Run now..., green + New task.

Jobs table (left, main) Columns: NAME | SCHEDULE | COMMAND | LAST | NEXT. Each row has a status dot on the left (green = active, grey = disabled). NEXT run is bold when upcoming. | | NAME | SCHEDULE (cron) | COMMAND | LAST | NEXT | | | | | | | | | | green
 | Nightly restart | 0 5 * * * | sv_shutdown | 2026-05-21 05:00 | **2026-05-22 05:00** | | green | Warmup reset | */30 * * * * | mp_warmup_start | 12:30 | **13:00** | | green | Workshop sync | 0 4 * * 0 | ds_workshop_sync | 2026-05-18 04:00 | **2026-05-25 04:00** | | green | Daily backup | 0 3 * * * | exec ./backups/snap.sh | 2026-05-21 03:00 | **2026-05-22 03:00** | | grey | Pug auto-shuffle | 0 19 * * 5 | mz_shuffle | 2026-05-16 19:00 | 2026-05-23 19:00 (disabled) |

RECENT RUNS table (below jobs) Columns: WHEN | TASK | DURATION | RESULT. RESULT is a colored pill: green ok, amber N retries, grey disabled. | WHEN | TASK | DURATION | RESULT | | | | |
 — | — | | 12:30:00 | Warmup reset | 0.08s | ok | | 12:00:00 | Warmup reset | 0.07s | ok | | 05:00:00 | Nightly restart | 14.2s | ok | | 03:00:00 | Daily backup | 2m 04s | ok | | 04:00 (Sun) | Workshop sync | 8m 41s | 3 retries (amber) | | 19:00 (Fri) | Pug auto-shuffle | — | disabled (grey) |

Presets panel (right) Header: **Presets**. Clickable preset cards (name + cron/trigger + command, monospace grey): - Nightly restart — 0 5 * * * → sv_shutdown - Warmup reset every 30m — */30 * * * * → mp_warmup_start - Pre-match script — on-event match → matchzy_load mr12 - GOTV demo cleanup — 0 6 * * 0 → find demos -mtime +30 -delete - Discord ping if down — on-event down → webhook discord_alerts

Next 24h panel (right, below presets) Header: **Next 24h**. Timeline list of upcoming runs (time + task): - 13:00 Warmup reset - 13:30 Warmup reset - 14:00 Warmup reset - - hourly until 04:30 - (collapsed range indicator) - 05:00 Nightly restart - 03:00 Daily backup

Implementation notes: cron scheduling runs via nokit-relay. Jobs support both cron expressions and on-event triggers (match, down). System timezone shown: Europe/Berlin.

10. Admin

Page title: **Admin**. Subtitle: 4 users · audit retention 90d · Discord webhook configured · GOTV port 27020. Top-right buttons: Export audit log, green + Invite admin.

Tabs Users & roles 4 · Audit log 10 · Integrations · GOTV & demos.

Users & roles tab Users table columns: **USER** | **EMAIL** | **ROLE** | **MFA** | **LAST SEEN**. **ROLE** is a colored pill; **MFA** is green **on** / amber **off** pill. | **USER** | **EMAIL** | **ROLE** | **MFA** | **LAST SEEN** | |——|——|——|——|——| | m0use@ | m@m0use.dev | owner (green) | on | 12:51 — now | | jaeger | p.j@axum.io | admin (blue) | on | 2h ago | | kira | kira@scrim.gg | admin (blue) | off (amber) | 1d ago | | ref-bot | discord://ref | moderator (grey) | on | 4d ago |

Role matrix panel (right) — Header **Role matrix**. Columns: **CAPABILITY** | **OWNER** | **ADMIN** | **MOD** (= granted, · = not): | **CAPABILITY** | **OWNER** | **ADMIN** | **MOD** | |———|:—:|:—:|:—:| | rcon console | | | · | | kick / ban | | | | edit configs | | | · | | install plugins | | | · | | manage admins | | · | · | | manage servers | | · | · |

Audit log tab Controls: search **search by user, action, target, time;** time-range segmented filter [**today** | **7d** | **30d** | **all**] (today selected); type filter [**all** | **writes** | **signin**] (writes selected). Columns: **WHEN** | **WHO** | **ACTION** | **TARGET** + a status check (ok / warning icon at far right). **ACTION** is a colored keyword pill (cmd, kick, plugin, map, preset, config, ban, signin). | **WHEN** | **WHO** | **ACTION** | **TARGET** | |——|——|——|——|——| | 12:51:08 | m0use@ | cmd | mp_warmup_end | | 12:49:51 | jaeger | kick | h4rdy_woods (STEAM_1:1:77124930) — afk | | 12:48:02 | system | plugin | WeaponPaints warn: fetch timeout | | 12:40:11 | m0use@ | map | de_mirage (was de_inferno) | | 12:38:00 | jaeger | preset | apply: MR12 (24+OT) | | 12:35:12 | m0use@ | config | edit gamemode_competitive.cfg (+4 -2) | | 12:30:00 | cron | cmd | mp_warmup_start | | 12:18:54 | system | plugin | GotvFix bind failure | | 12:14:33 | kira | ban | lobotomy — wallhack — 21d | | 12:02:11 | m0use@ | signin | from 84.124.221.9 — chrome/macOS | |

Integrations tab (4 cards) **Discord webhook** — status **connected** (green): - url <https://discord.com/api/webhooks/123...../k83a1.....> - channel #nokit-fra1 - events match start · match end · server down · plugin error · ban - last 12:51 - match started - Buttons: test message, edit, red disconnect.

Steam workshop API — status **key valid** (green): - key A1F8.....E0 - quota 4,221 / 100,000 calls (24h) - collection ws://3142099912 - 16 maps - last sync 2026-05-18 04:08 · 8m 41s · 0 errors - Buttons: rotate key, green sync now.

Steam game-server login token — status **active** (green): - app 730 (cs2) - token F38C1B..... - label nokit-fra1 - added 2026-04-11

Backup target — status **S3:** - bucket s3://nokit-backups - prefix fra1.mm/ - retention 30d (cfg) · 14d (demos) - last 03:00 · 412 MB · ok

GOTV & demos tab Demo files table columns: **FILE** | **SIZE** | **DURATION** | **DLS** (downloads): | **FILE** | **SIZE** | **DURATION** | **DLS** | |——|——|——|

— —		2026-05-21_de_mirage_mr12_team-0_vs_team-B.dem		94.2 MB		46m		2		2026-05-20_de_inferno_scrim_axum_vs_nbk.dem		102.8 MB		52m		0		2026-05-20_de_nuke_pug.dem		78.4 MB		38m		1		2026-05-19_de_anubis_eu-quals.dem		121.0 MB		1h 04m		8	
-----	--	--	--	---------	--	-----	--	---	--	---	--	----------	--	-----	--	---	--	----------------------------	--	---------	--	-----	--	---	--	-----------------------------------	--	----------	--	--------	--	---	--

GOTV settings panel (right) — Header **GOTV settings** (cvar → value):
- tv_enable 1 - tv_port 27020 - tv_maxclients 32 - tv_delay 90s -
tv_autorecord 1 - tv_chatgroup "nokit-gotv" - status 1 spectator
· 0.4 mbps (green) - **DEMO RETENTION**: keep [14 days] then [upload to S3] (editable pills).

11. Multi-server switcher

Opened from the top-left server dropdown (fra1.mm · 5.39.42.118:27015).
Dropdown panel: - Section label **SERVERS**. Each row: online status dot + name + ip:port · region + player-count badge + × remove button. | Name | Address | Region | Players | | | | | | | fra1.mm (active) | 5.39.42.118:27015 | Frankfurt | 9/12 | | seal.scrim | 10.0.1.22:27015 | Seattle | 10/10 | | home.lan | 192.168.1.40:27015 | Local | 3/10 | | practice-aws | 54.218.7.92:27015 | us-west-2 | 0/8 | | retake-old | 88.99.10.4:27015 | Hetzner | — (offline, grey dot) | - Section label **ADD SERVER** — inline form: name text input, ip:port text input, green **Add** button, and a rcon password input below (full-width). - Clicking a server row switches the active server (updates top-bar tickrate/players/map and all pages). Corresponds to CLI `srv add [hostname] [IP:PORT]`.

12. Login / Auth & Session

The public demo (nokit.app/demo) is a pre-authenticated single-page mock — there is **no separate /login route** (visiting /login returns 404). Auth is documented rather than shown as a live form: - **No SaaS accounts / no telemetry**. Self-hosted only. - Initial admin set via env var in docker-compose: `NOKIT_ADMIN: m0use@local`. - Authentication is via **SteamID or email**, with **MFA support** (see Admin → Users & roles: MFA on/off per user). - **Role-based access control**: Owner / Admin / Moderator, enforced by the per-capability role matrix (see Admin). - Sign-in events are recorded in the Audit log (e.g. `signin` - from 84.124.221.9 - chrome/macOS). - Current session identity is shown in the top bar as m0use@ with an avatar; a **power/logout icon** sits at the far right of the top bar. (In the demo these are static — the user menu does not open a dropdown.) - Session/connection state surfaced globally: sidebar footer `WS` indicator (note: docs describe live log transport as **SSE**, and RCON over loopback; the UI also shows a WS status pill) and footer `srcds OK · relay OK`.

This fork's implementation note: Auth is currently password-based login with **session tokens persisted in SQLite** (see Feature Status Table §1).

13. Global search / Command palette

- Top-bar search field placeholder `cvars, players...` with a `K` keyboard shortcut badge.
 - Intended as a command palette to jump to `cvars/players/pages`. In the demo, activating it routes to the **RCON Console** (the `K` shortcut and the sidebar RCON `K` badge are linked). No floating overlay palette renders in the static demo.
-

Cross-cutting UI / Design notes (for re-implementation)

- **Theme:** dark UI (near-black background `#0d0d0f`-ish), green primary accent (buttons, active nav, online dots), monospace fonts for logs/console/cvars, sans-serif for chrome.
- **Status color language:**
 - **green** = ok / online / enabled / active
 - **amber/orange** = warning / T-side / update-available / temporary
 - **red** = error / failed / ban / danger / permanent
 - **grey** = disabled / offline
 - **blue** = CT-side / admin-role / core-plugin / info
- **Common layout pattern:** page header (title + descriptive subtitle on the left, action buttons on the right) → optional tab bar with count badges → main content (often a left main panel + right sidebar of secondary panels).
- **Segmented toggles** used widely (match/practice/pug, all/CT/T/spec, live/paused, time ranges).
- **Tables** use uppercase small-caps column headers, colored pill badges for status/role/team, and reveal inline row-action buttons on hover/selection.
- **Every mutating action** is designed to be logged to the Audit log with actor + action + target.
- **Refresh cadence:** dashboard/stats and player list auto-refresh every ~4s; logs stream via SSE.
- **Update frequency example:** page footer clock ticks live (observed 16:16 → 16:30 across the session), timezone **Europe/Berlin** (UTC+2).
- **Version strings:** app `nokit v0.1.0-rc.1`, relay `v0.4.2`, CS# `1.0.305`, Metamod `2.0.0-12`, CS2 build `1.40.6.3` / protocol `v15 build 9462`.
- **Deploy target:** Docker image `ghcr.io/nokit/nokit:latest`, default web port 8080, data volume `/data`, runs alongside `srcds` (loopback RCON). Single ~14 MB Go binary.

This document was compiled from observations of the nokit demo. Keep it updated as features move from (demo/planned) to (implemented).